

38.4 Overdispersion

The word has been used loosely so far. This section says where the extra variability comes from, how to see it, what ignoring it costs, and which repair to make.

38.4.1 Where it comes from

Proposition 38.4.1 — Sources of overdispersion, and the price of ignoring it

- (a) *Unobserved heterogeneity.* Let $Y \mid \Lambda$ have mean Λ and variance $V^*(\Lambda)$, and let Λ have mean μ and variance τ^2 , with $0 < \tau^2 < \infty$ and $E\{V^*(\Lambda)\} < \infty$. Then $EY = \mu$ and $\text{Var } Y = E\{V^*(\Lambda)\} + \tau^2$, which for a Poisson kernel $V^*(\lambda) = \lambda$ is $\mu + \tau^2 > \mu$: any unmodelled variation in the mean overdispersed.
- (b) *Clustering.* If the response is an average of m units with pairwise correlation $\rho > 0$, its variance is inflated by $1 + \rho(m - 1)$, by Lemma 38.1.1; the same factor applies to a sum of m correlated indicators.
- (c) *Contagion.* Let $Y = \sum_{j=1}^N S_j$, where N is Poisson with mean ν and the S_j are independent copies of a positive integer-valued S with $ES^2 < \infty$, independent of N : events arrive in clusters. Then $EY = \nu ES$ and $\text{Var } Y = \nu ES^2$, so

$$\frac{\text{Var } Y}{EY} = \frac{ES^2}{ES} \geq ES \geq 1, \quad (38.4.1)$$

with equality throughout if and only if $S \equiv 1$, that is unless the events arrive singly.

- (d) *Neither of the above.* A wrong mean model, an omitted interaction or a few gross outliers all inflate X^2 , and no dispersion parameter repairs them; for ungrouped binary data a large X^2 is always a statement about the mean model.
- (e) *The price.* Suppose the mean model is correct and the true variances are $\sigma_i^2 = \phi_0 V(\mu_i^0)/w_i$ with $\phi_0 \geq 1$ unknown, while inference uses $\phi = 1$. Then under (Q1)–(Q5) any statistic of Proposition 34.5.3 for a q -dimensional hypothesis converges to $\phi_0 \chi^2(q)$, so a nominal level- α test has asymptotic size

$$\Pr\{\chi^2(q) > \chi_{1-\alpha}^2(q)/\phi_0\} \geq \alpha, \quad (38.4.2)$$

increasing in ϕ_0 , and tending to 1 as $q \rightarrow \infty$ for every fixed $\phi_0 > 1$. Standard errors are too small by the factor $\phi_0^{1/2}$.

Proof.

- (a) The law of total variance gives $\text{Var } Y = E\{V^*(\Lambda)\} + \tau^2$, and $E\Lambda = \mu$; this is Lemma 37.4.1(a) with the mixing distribution left arbitrary.
- (b) is Lemma 38.1.1.
- (c) Conditioning on N gives $E(Y \mid N) = N ES$ and $\text{Var}(Y \mid N) = N \text{Var } S$, so

$EY = \nu ES$ and

$$\begin{aligned}\text{Var } Y &= E(N) \text{Var } S + \text{Var}(N)(ES)^2 \\ &= \nu \{ \text{Var } S + (ES)^2 \} = \nu ES^2,\end{aligned}$$

using $\text{Var } N = EN = \nu$. Dividing gives $\text{Var } Y / EY = ES^2 / ES$; the first inequality in Eq. 38.4.1 is $ES^2 \geq (ES)^2$, equality only for degenerate S , and the second is $ES \geq 1$. Equality in both forces $S \equiv 1$, and then $Y = N$ is Poisson.

- (d) The binary claim is the warning of Section 38.1; the rest records what X^2 measures and is not a mathematical assertion.
- (e) By Theorem 38.3.1(a) with $\mathbf{B}_n = \phi_0 \mathbf{A}_n$ — which is Theorem 38.2.2(e) at $\sigma_i^2 = \phi_0 V_i / w_i$ with the working $\phi = 1$ — $\mathbf{A}_n^{1/2}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^0) \rightarrow N_p(\mathbf{0}, \phi_0 \mathbf{I})$. The Wald statistic computed as though the covariance were $\mathbf{A}_n^{-1}$ is $\|(\mathbf{I} - \mathbf{P}_n) \mathbf{A}_n^{1/2}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^0)\|^2$ for a projection $\mathbf{P}_n$ of rank $p - q$, as in the proof of Proposition 34.5.3, hence converges to ϕ_0 times a $\chi^2(q)$ variable; the other two differ from it by $o_p(1)$ by the same proof, since Proposition 38.3.2(c) extends the expansion to Q . The size Eq. 38.4.2 follows and exceeds α because $\phi_0 > 1$ lowers the threshold; the limit as $q \rightarrow \infty$ is Exercise (A1).

■

These sizes are worse than intuition suggests. At $\phi_0 = 2$ a nominal 5% test of one coefficient rejects with probability 0.166 and of six coefficients with probability 0.391; at $\phi_0 = 5$ the figures are 0.381 and 0.866. A dispersion of five is not unusual — the strike data of Example (Strike durations) have $\hat{\phi}_P = 39.89$.

```
import numpy as np
from scipy import stats

def naive_size(phi, q):
    """Size of a nominal 5% chi-squared test when the statistic is inflated by phi.
    """
    return stats.chi2.sf(stats.chi2.ppf(0.95, q) / phi, q)

print("phi    q = 1    q = 3    q = 6")
for phi in (1.5, 2.0, 3.0, 5.0, 10.0):
    print(f"{phi:4.1f} " + " ".join(f"{naive_size(phi, q):7.3f}" for q in (1, 3, 6))
    ))
```

[Run this code online](#)

Example. (A test that rejects six times too often)

Counts were simulated with a constant mean 6.0, a variance 4.0 times the mean and a true slope of zero, and a Poisson log-linear model was fitted to each of 5000 data sets of size 100. Referring the drop in deviance to $\chi^2(1)$ rejects at the nominal 5% level in 0.322 of the replicates — Eq. 38.4.2 predicts 0.327 at $\phi_0 = 4$ — with the statistic averaging 3.90 against a $\chi^2(1)$ mean of one. Dividing by $\hat{\phi}_P$ and referring to $F(1, 98)$ rejects in 0.051, and the sandwich Wald test Eq. 38.3.2 in 0.058 (Figure 38.4.1).

```
def poisson_fit(X, y, steps=20):
    beta = np.zeros(X.shape[1])
    beta[0] = np.log(max(y.mean(), 0.5))
```

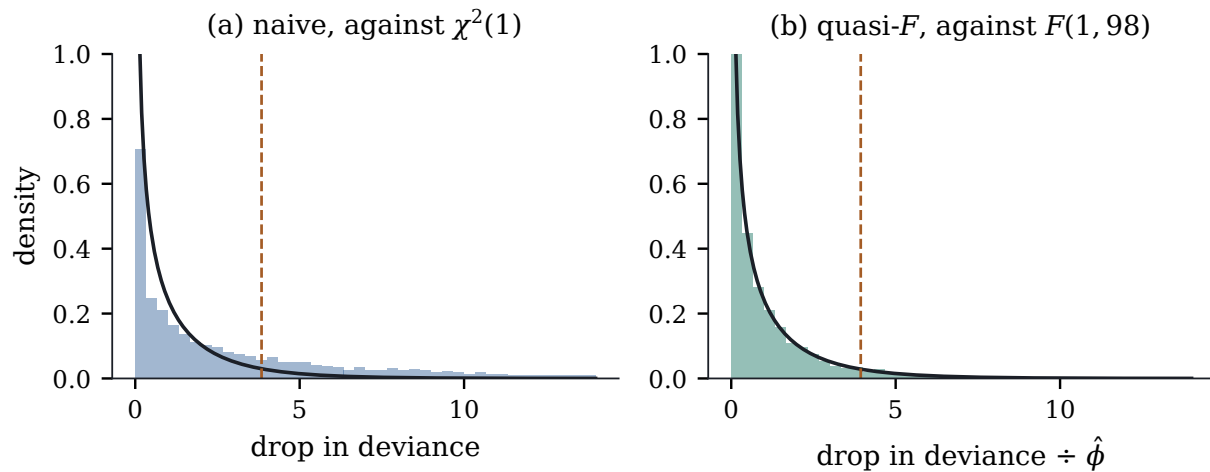


Figure 38.4.1: Simulated null distributions of the drop in deviance for one coefficient, on data whose variance is four times their mean: (a) the raw statistic against $\chi^2(1)$; (b) the statistic divided by the Pearson dispersion, against $F(1, 98)$. The dashed line is the upper 5% point.

```

for _ in range(steps):
    mu = np.exp(X @ beta)
    z = X @ beta + (y - mu) / mu
    beta = np.linalg.solve((X.T * mu) @ X, (X.T * mu) @ z)
return beta, np.exp(X @ beta)

def poisson_deviance(y, mu):
    return 2 * np.sum(np.where(y > 0, y * np.log(np.maximum(y, 1e-300) / mu), 0.0) -
                      (y - mu))

def null_statistics(n, phi, mu0, reps, rng):
    """Three statistics for the slope, on data whose true slope is zero."""
    x = np.linspace(-1.0, 1.0, n)
    X = np.column_stack([np.ones(n), x])
    shape = np.full(n, mu0 / (phi - 1.0)) # negative binomial with
    variance phi*mu
    out = np.empty((reps, 3))
    for r in range(reps):
        y = rng.negative_binomial(shape, shape / (shape + mu0)).astype(float)
        beta, mu = poisson_fit(X, y)
        beta0, mu_null = poisson_fit(X[:, :1], y)
        drop = poisson_deviance(y, mu_null) - poisson_deviance(y, mu)
        phi_hat = np.sum((y - mu) ** 2 / mu) / (n - 2)
        bread = np.linalg.inv((X.T * mu) @ X)
        sand = bread @ ((X * ((y - mu) ** 2)[:, None]).T @ X) @ bread
        out[r] = [drop, drop / phi_hat, beta[1] ** 2 / sand[1, 1]]
    return out

N, PHI, MU0 = 100, 4.0, 6.0
rng = np.random.default_rng(3841)
stat = null_statistics(N, PHI, MU0, 400, rng)
rates = [np.mean(stat[:, 0] > stats.chi2.ppf(0.95, 1)),
         np.mean(stat[:, 1] > stats.f.ppf(0.95, 1, N - 2)),
         np.mean(stat[:, 2] > stats.chi2.ppf(0.95, 1))]
print(f"rejection rates at the 5% level: naive {rates[0]:.3f},"
      f" quasi-F {rates[1]:.3f}, sandwich Wald {rates[2]:.3f}")

```

[Run this code online](#)

38.4.2 Seeing it

Four diagnostics, in the order in which they should be used. **Check the mean model first:** an omitted term can remove an apparent dispersion entirely, as Example (Twenty thousand records in thirty-two rows) showed, and only a mean model that has survived Definition 34.5.2 and the plots of [Chapter 20](#) makes a dispersion parameter mean what it says. **Compare $\hat{\phi}_P = X^2/(n-p)$ with one,** remembering that it is calibrated only where Theorem 34.5.5(b) applies, noisy for ungrouped counts with small means, and meaningless for ungrouped binary data. **Use a score test:** for counts, Theorem 37.4.3 tests the Poisson against NB2 alternatives from the Poisson fit alone, with a one-sided alternative. **Compare the two covariances:** a large discrepancy between $\hat{\phi}_P(\mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X})^{-1}$ and the sandwich Eq. 38.3.2 says the *shape* of the variance function is wrong and not merely its scale, while a small one, as in Example (Four standard errors for the strike data), is evidence that the shape is right (Exercise (C1)). A plot of squared Pearson residuals against fitted means, and against group sizes where they vary, shows the same in more detail.

38.4.3 Choosing a repair

Four devices are in use, differing in what they assume and what they give.

device	assumes	gives	does not give
quasi model with $\hat{\phi}$	mean and variance shape	$\hat{\beta}$, F tests, model-based errors	likelihood, AIC, predictions
quasi model with the sandwich	mean model only	$\hat{\beta}$, Wald tests	efficiency, small samples
NB2, NB1 or beta-binomial	a full distribution	likelihood, AIC, predictions	robustness to that choice
random effect	a full distribution and a mixing law	conditional coefficients, prediction of units	the marginal coefficients

Putting a dispersion parameter into a quasi model changes the standard errors and leaves $\hat{\beta}$ exactly where it was (Theorem 38.2.2(a)); changing to a different likelihood with the same variance function can move it by more than a standard error ([Example \(Four analyses of the strike data\)](#)). The fourth device changes the estimand rather than the estimate: a coefficient conditional on a unit's own random effect answers a different question from one describing a population average, a distinction Chapter 40 takes up and does not treat as small.

Example. (Four analyses of the strike data)

Continuing Example (Strike durations) and Example (Four standard errors for the strike data), the production coefficient under four treatments of the extra variability is

fit	coefficient	standard error
Poisson, $\phi = 1$	-7.680	0.406
quasi-Poisson, $V = \phi\mu$	-7.680	2.564
NB1, $\text{Var} = \phi\mu$	-4.934	1.983
NB2, $\text{Var} = \mu + \mu^2/\kappa$	-9.327	2.931

The NB1 fit has $\hat{\phi} = 42.78$, close to the quasi-Poisson $\hat{\phi}_P = 39.89$, and *the same variance function*; yet its coefficient differs by more than a standard error, because its estimating equation Eq. 37.4.6 is not Eq. 38.1.1 — the warning of [Section 38.1](#) in numbers. The NB2 fit, with $\hat{\alpha} = 1/\hat{\kappa} = 0.954$, gives -9.327 against the -9.353 of the quasi model with $V(\mu) = \mu^2$: at these fitted means the quadratic term of the NB2 variance dominates, so the two nearly coincide.

The tests agree on the conclusion and not on its strength. The drop in deviance is 350.9, overwhelming as $\chi^2(1)$; the quasi- F statistic Eq. 38.3.3 is 8.80 on (1, 60) degrees of freedom, $p = 0.0043$, and the score test for overdispersion Eq. 37.4.8 is 224.5. Unanticipated production predicts shorter strikes, at about the evidence a t statistic of three provides; the Poisson analysis overstates that by orders of magnitude.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

data = sm.datasets.strikes.load_pandas().data
y = data["duration"].to_numpy(float)          # length of the strike, in days
x = data["iprod"].to_numpy(float)             # unanticipated industrial production
X = np.column_stack([np.ones(len(y)), x])
n, p = X.shape

def quasi_irls(X, y, V, steps=60):
    """Solve the quasi-score equations for a log link and variance function V."""
    beta = np.zeros(X.shape[1])
    beta[0] = np.log(y.mean())
    for _ in range(steps):
        eta = X @ beta
        mu = np.exp(eta)
        W = mu ** 2 / V(mu)                    # working weight (dmu/deta)^2 / V(mu)
        z = eta + (y - mu) / mu                # working response
        beta = np.linalg.solve((X.T * W) @ X, (X.T * W) @ z)
    return beta

beta_lin = quasi_irls(X, y, lambda m: m)      # V(mu) = mu
mu = np.exp(X @ beta_lin)
deviance = 2 * np.sum(y * np.log(y / mu) - (y - mu))  # quasi-deviance, V(mu)
              = mu
pearson = np.sum((y - mu) ** 2 / mu)          # X^2 with V(mu) = mu
phi_pearson = pearson / (n - p)
beta_null = quasi_irls(X[:, :1], y, lambda m: m)
mu_null = np.exp(X[:, :1] @ beta_null)
deviance_null = 2 * np.sum(y * np.log(y / mu_null) - (y - mu_null))
naive_chisq = deviance_null - deviance        # the Poisson likelihood
ratio
quasi_f = (deviance_null - deviance) / phi_pearson  # referred to F(1, n - p)
print(f"naive chi-squared {naive_chisq:.1f}, p = {stats.chi2.sf(naive_chisq, 1):.2e}")
```

```

    ")
print(f"quasi F {quasi_f:.2f} on (1, {n - p}) df, p = {stats.f.sf(quasi_f, 1, n - p)
      :.4f}")

```

[Run this code online](#)

Idea. The quasi-score does not know about the variance at all (Theorem 38.2.2(b)), so a wrong variance costs efficiency and calibration but not consistency. Whenever a colleague reports that “adding a dispersion parameter did not change the results”, the coefficients were being looked at and the standard errors were not.

38.4.4 Exercises

38.4.4.1 A. Check your understanding

Exercise A1.

Show that for fixed $\phi_0 > 1$ the size [Eq. 38.4.2](#) tends to 1 as $q \rightarrow \infty$. (Use $\chi^2(q)/q \rightarrow 1$ in probability, together with $\chi^2_{1-\alpha}(q)/q \rightarrow 1$, and compare with the event $\chi^2(q)/q > \chi^2_{1-\alpha}(q)/(q\phi_0)$.)

Solution. By the law of large numbers $\chi^2(q)/q \rightarrow 1$ in probability, and the same concentration gives $\chi^2_{1-\alpha}(q)/q = c_q \rightarrow 1$. The event in [Eq. 38.4.2](#) is $\{\chi^2(q)/q > c_q/\phi_0\}$, whose threshold tends to $\phi_0^{-1} < 1$, so its probability tends to 1; at $q = 1$ it is the finite $\Pr\{\chi^2(1) > 3.841/\phi_0\}$. The listing of this section shows the size increasing with q over the range tabulated, but that is a computation, not a proof.

Exercise A2.

A Poisson regression of accident counts on three covariates gives $X^2/(n-p) = 3.2$. Give three distinct explanations, and for each one say what you would compute next.

38.4.4.2 B. Practice

Exercise B1.

In [Proposition 38.4.1\(c\)](#), let S be geometric on $\{1, 2, \dots\}$ with mean $1/\theta$. Compute $\text{Var } Y / \text{E } Y$ and show that it equals $(2 - \theta)/\theta$. For which θ does the compound Poisson have the variance-to-mean ratio of a negative binomial with $\kappa = \mu$?

Exercise B2.

Grouped binomial data have group sizes m_i spread between 2 and 50. Show that under the constant-dispersion specification the Pearson terms $m_i(y_i - \hat{\mu}_i)^2 / \{\hat{\mu}_i(1 - \hat{\mu}_i)\}$ have expectation ϕ free of m_i , while under [Eq. 38.1.3](#) they have expectation $1 + \rho(m_i - 1)$. Describe the plot this suggests and what each specification predicts for it.

Solution. Under $\text{Var}(Y_i) = \phi\mu_i(1 - \mu_i)/m_i$ the Pearson term at the true mean has expectation ϕ ; under Eq. 38.1.3 the same calculation gives $1 + \rho(m_i - 1)$, linear in m_i with slope ρ . So plot squared Pearson residuals against m_i with a smooth: flat supports a constant dispersion, increasing the exchangeable form.

Exercise B3.

Both the NB1 model and the quasi-Poisson specification have $\text{Var}(Y_i) = \phi\mu_i$. Using Eq. 37.4.6 and Eq. 38.1.1, explain why a single very large count influences the two fits differently, and predict which of the two coefficients in [Example \(Four analyses of the strike data\)](#) would move more if the longest strike were deleted.

Solution. In the quasi-Poisson equation the log link cancels μ_i against $V(\mu_i)$, so observation i enters through the raw residual $y_i - \mu_i$: the longest strike, a count of 216 against a fitted mean of 58.99, contributes 157.0. The NB1 score Eq. 37.4.6 contains y_i only inside $\psi(y_i + \alpha\mu_i)$, which grows like $\log y_i$, so the NB1 fit is far less sensitive to it and the quasi-Poisson coefficient should move more when it is removed.